KATHMANDU UNIVERSITY
Department of Computer Science and Engineering

A Project Defense Presentation on

"Readify"

A Social Book Recommendation Platform using Graph Attention Networks (GAT)

PREPARED BY

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3rd Year / 2nd Semester

Agendas

- 1. Introduction
  Background, objectives & motivation
- 2. Related Works
  Existing platforms & GAT literature
- 3. System Design
  Architecture, database & recommendation engine
- 4. Requirements & Planning
  Specs, UI, timeline, conclusion

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01

Introduction

Setting the stage: why Readify, and what it aims to achieve

Background & Problem Statement

Readers increasingly rely on online communities (Goodreads, StoryGraph, Hardcover) to discover books — but most recommendation engines behind them are limited.

- **Cold-Start Problem**

  New users/books get poor recommendations

- **Data Sparsity**

  Most user–book interactions are missing

- **Popularity Bias**

  Only popular titles keep getting recommended

- **Weak Relationship Modelling**

  Social & complex relations are ignored

Our Approach

Model users, books, ratings, reviews & social ties as a graph — then apply Graph Attention Networks (GAT) so the model learns which neighbours matter most.

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Project Objectives

To design and develop an intelligent social book recommendation platform that delivers personalised recommendations using Graph Attention Networks while fostering an interactive reading community.

- **Design & Develop**

A web-based social book recommendation platform named Readify.

- **Construct a Graph**

Represent users, books, ratings, reviews, genres & social relationships.

- **Build the Engine**

Implement an attention-based recommendation engine using GAT.

- **Evaluate & Compare**

Assess the model against conventional recommendation approaches.

02

Existing Systems and Platforms

Collaborative Filtering

Recommends based on similarity among users or items (user-based / item-based).

LIMITATION

Cold-start, sparse matrices, popularity bias

Content-Based Filtering

Uses genre, author, keywords & description similarity to prior likes.

LIMITATION

Limited diversity — repeats similar attributes

Hybrid Systems

Combine collaborative & content-based methods to offset individual weaknesses.

LIMITATION

Still struggles with complex social relationships

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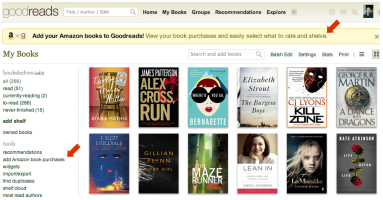
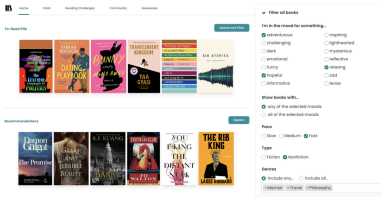
Existing Book Recommendation Platforms



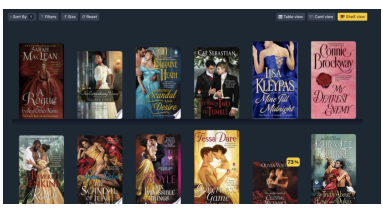
Goodreads

Collaborative filtering + interaction history; static, no graph learning.



StoryGraph

Mood/pace/theme metadata-driven; no deep graph model.



Hardcover

Strong social features; traditional recommendation logic only.

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Graph Neural Networks & Graph Attention

GNNs learn from both node features AND relationships between nodes — ideal when users, books, authors, genres & reviews naturally form a graph.

- GAT (Veličković et al., 2018) assigns unequal importance to neighbouring nodes via an attention mechanism.
- Instead of treating every neighbour equally, GAT learns attention coefficients per edge.
- This lets the model focus on highly relevant interactions and suppress noisy ones.
- Applied successfully in recommendation, fraud detection, social & biological network analysis.

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Comparison of Existing Systems & Research Gap

System	Social	AI Recs	Graph Learning	Major Limitation
Goodreads	Yes	Yes	No	Popularity bias, traditional methods
StoryGraph	Yes	Partial	No	Limited use of social relationships
Hardcover	Yes	Partial	No	Comparatively simple recommender
Readify	Yes	Yes	Yes (GAT)	

- Research Gap

Existing platforms rarely use adaptive attention to identify influential readers, so recommendations overlook trusted reviewer signals and indirect community interactions. Readify closes this gap by applying GAT over a unified user–book social graph.

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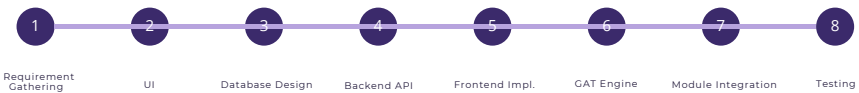
03

System Design & Methodology

Development Methodology

Readify followed an iterative software development methodology, continuously refined across phases:



Three-Tier Architecture

Presentation (React.js) → Application (Node.js/Express) → Data (PostgreSQL) — with a Python/PyTorch recommendation service alongside the backend.

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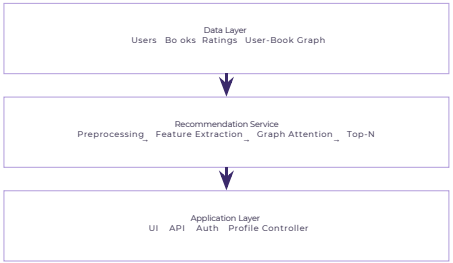
Overall System Architecture



Layers

Data Layer

Users, books, ratings, user-book graph

Recommendation Service

Preprocessing → features → graph attention → Top-N

Application Layer

UI, API, Auth, profile controller

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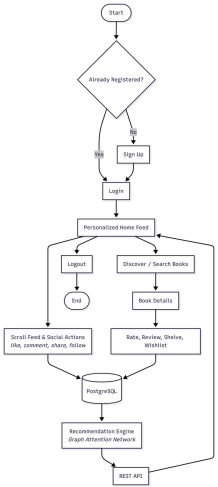
System Workflow



- User registers/logs in (OTP or Google OAuth). Browses,
- searches, rates, reviews, shelves & follows readers.
  Every interaction is written to PostgreSQL. Interactions
- continuously update the user-book graph. GAT
- aggregates neighbour info & computes attention
- weights.
  Top-N recommendations return via the REST API to the
- home feed.

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Technology Stack

- **Frontend**
  React.js
- **Backend**
  Node.js + Express.js
- **Database**
  PostgreSQL
- **Authentication**
  JWT + Google OAuth

Recommendation Engine
Python

ML Framework
PyTorch + PyG

Version Control
Git & GitHub

Design & Testing
Figma + Postman

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Database Design

Relational schema in PostgreSQL — Users, Books, Authors, Genres, Ratings, Reviews, Followers, Reading Lists & Recommendations, linked via primary/foreign keys.

- Users
- Books
- Authors
- Genres
- Ratings
- Reviews
- Followers
- Reading Lists
- Recommendations

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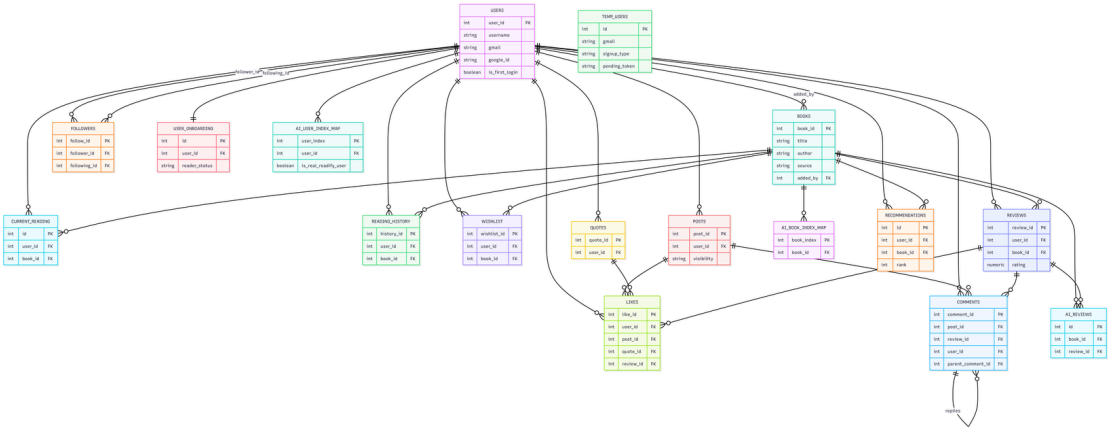
Database Design



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Recommendation Engine & Pipeline

The graph combines User–Book interactions, User–User follows, reviews, genres, authors & ratings. Nodes carry feature vectors; Graph Attention layers aggregate neighbour information with multiple attention heads for robustness.

Multiple attention heads improve representationlearning and recommendation robustness.

Recommendation Pipeline:

1. User performs interactions
2. Interaction data stored in PostgreSQL
3. Graph data constructed
4. Node features generated
5. GATcomputesembeddings
6. Compare user & candidate book embeddings
7. Top-N recommendations generated
8. Recommendations displayed on homepage

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Authentication Process

Registration → Verification (OTP / OAuth) → Password Encrypted → JWT Issued → Session Managed

OTP Verification

Manual registration verified via one-time password.

Google OAuth 2.0

One-click sign-in delegating identity verification.

Bcrypt Encryption

Passwords securely hashed before storage.

JWT Sessions

Tokkens issued to manage sessions securely.

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04

Requirements & Conclusions

Functional & Non-Functional Requirements

FUNCTIONAL

- Registration (manual + Google Sign-In) & secure authentication
- Book search, browsing & detailed book pages
- 5-star ratings and full review system
- Multiple bookshelves — Reading, Completed, Wishlist, Custom
- Social features: follow, likes, comments, activity feed GAT-based personalised recommendations & notifications
- Admin controls for users, books & reported content

NON-FUNCTIONAL

- High availability & fast recommendation response
- Secure authentication and authorization
- Scalable, RESTful backend architecture
- Responsive, cross-platform user interface
- Reliable database transactions
- Secure password encryption
- Easy maintenance & future extensibility

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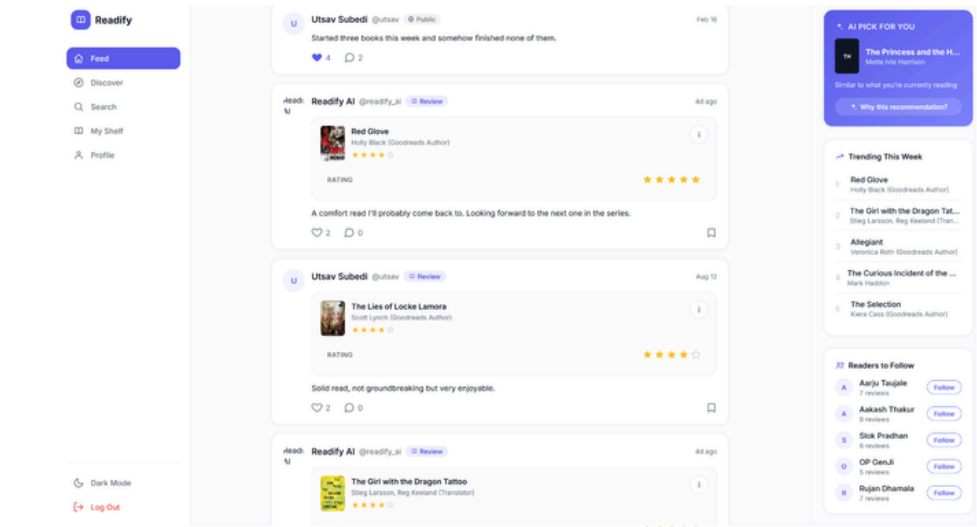
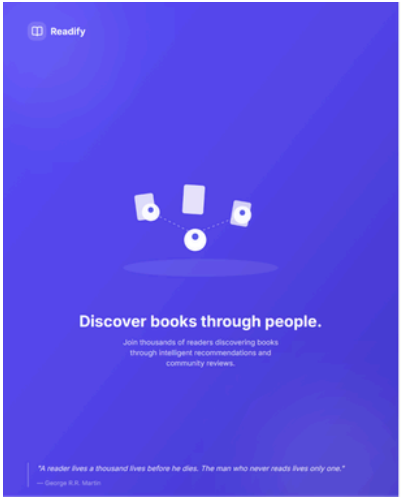
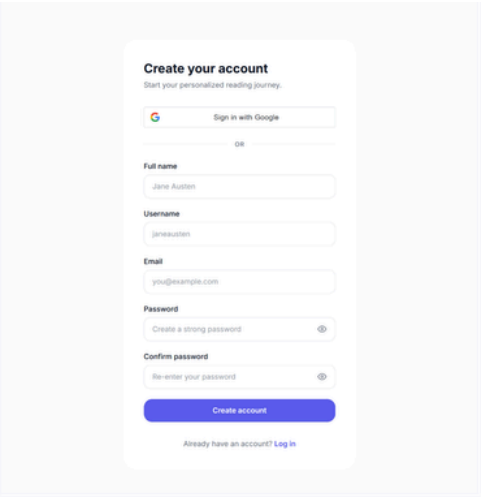
User Interface Design



User Interface Design





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Conclusion & Limitations

CONCLUSION

Readify combines social networking with Graph Attention Networks to model users, books, ratings, reviews and social ties as a connected graph — capturing complex relationships that conventional collaborative or content-based filtering miss, while encouraging real community engagement.

LIMITATIONS

- Recommendation quality depends on sufficient interaction data
- GAT training requires GPU resources for efficiency New users get limited recommendations until history builds up
- Model evaluated on a limited dataset.

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Future Work

- **LLM Integration**
  Conversational, natural-language book recommendations.
- **Mobile App**
  Native experience built with Flutter.
- **Cloud-Scale Deployment**
  Production-grade infrastructure with multimodal recommendation.

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Thank You!

Questions & Discussion

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